

Measuring Clustering Quality

- ❑ Clustering Evaluation: Evaluating how good the clustering results are
 - ❑ No commonly recognized best suitable measure in practice
- ❑ Extrinsic vs. intrinsic methods: depending on whether ground truth is used
 - ❑ Ground truth: the ideal clustering built by using human experts
- ❑ Extrinsic: Supervised, employ criteria not inherent to the dataset
 - ❑ Compare a clustering against prior or expert-specified knowledge (i.e., the ground truth) using certain clustering quality measure
- ❑ Intrinsic: Unsupervised, criteria derived from data itself
 - ❑ Evaluate the goodness of a clustering by considering how well the clusters are separated and how compact the clusters are (e.g., silhouette coefficient)

General Criteria for Measuring Clustering Quality with Extrinsic Methods

- ❑ Given the ground truth C_g , $Q(C, C_g)$ is the quality measure for a clustering C
- ❑ $Q(C, C_g)$ is good if it satisfies the following four essential criteria
 - ❑ Cluster homogeneity: the purer, the better
 - ❑ Cluster completeness: assign objects belonging to the same category in the ground truth to the same cluster
 - ❑ Rag bag better than alien: putting a heterogeneous object into a pure cluster should be penalized more than putting it into a rag bag (i.e., “miscellaneous” or “other” category)
 - ❑ Small cluster preservation: splitting a small category into pieces is more harmful than splitting a large category into pieces

